Pietro Piccione

Résumé

pietropic@pm.me
pietropic.github.io/

Work experience

2025 – 2027 **Postdoc**, *University of Gothenburg*, Göteborg, Sweden, Supervisor: David Witt Nyström

Education

2022 – 2025 **PhD in Mathematics**, *Institut des mathématiques de Jussieu*, Sorbonne Université, Paris, France
Advisors: Sébastien Boucksom and Tat Dat Tô

2021 – 2022 **Master's degree in Mathematics**, *Institut des mathématiques de Jussieu*, Sorbonne Université, Paris, France

2017 – 2021 **Bachelor's degree in Mathematics**, *Instituto de Matemática e Estatística*, Universidade de São Paulo, São Paulo, Brazil
Prize for outstanding performance.

Research projects and grants

2026 – 2027 **Research grant: "Kähler and non-Archimedean geometry"**, *Funded by Stiftelsen Wilhelm och Martina Lundgrens Vetenskapsfond*
Grant of 50.000 SEK for research purposes.

2022 – 2025 **MathInParis2020**, *Scholarship for a PhD in Paris*, Cofunded by Marie Sklodowska-Curie Actions, Fondation sciences mathématiques de Paris

2021 – 2022 **PGSM Scholarship**, *Scholarship for a Master's degree in Paris*, Fondation sciences mathématiques de Paris

Articles

2025 **A transcendental non-Archimedean Calabi–Yau Theorem with applications to the cscK problem**, *joint with David Witt Nyström*, preprint, arXiv:2509.09442

2024 **A non-Archimedean theory of complex spaces and the csc-K problem**, Advances in Mathematics

Professional

Scientific Visits

2025 **Simons Semester in Analysis and Geometry on Complex Manifolds**, *Erdős Center*, Budapest
Three months visit to the program.

2024 **Special Geometric Structures and Analysis**, *SLMath*, Berkeley
Two months visit to the program.

Academic Services

2025 **Participated at the cycle "Un texte, une aventure mathématique"**, *Lycée international St Germain en Laye*, Slides and material
Talked at a high school in St Germain en Laye

2021 **Organizer of the A5 Group**, Instituto de Matemática e Estatística, Universidade de São Paulo
A student runned academic group with over 25 weekly seminars organized

2018 – 2019 **Deputy Student Representative in the Mathematics Department Council**, Instituto de Matemática e Estatística, Universidade de São Paulo, Brazil

Teaching

2026 **Riemann Surfaces**, *PhD course*, U. of Gothenburg and Chalmers University of Technology, Sweden

Teaching Assistantships

2025 **Introduction to Riemann Surfaces**, *Master's course*, Sorbonne Université, France
2024 **Topology and Differentiable Calculus I**, *Second year bachelor's course*, Sorbonne Université, France
2024 **Functional Analysis**, *Third year bachelor's course*, Sorbonne Université, France
2023 **Topology and Differentiable Calculus I**, *Second year bacherlor's course*, Sorbonne Université, France

Events Organized

2026 **Geometry and Topology of (Almost) Complex Structures**, *Online conference*
2022 **MaThematical Days : Paul Erdös**, *Universidade de São Paulo*, São Paulo, Brazil
2021 **MaThematical Days : Emmy Noether**, *Universidade de São Paulo*, São Paulo, Brazil
2021 **MaThematical Days : Élie Cartan**, *Universidade de São Paulo*, São Paulo, Brazil

Talks

A non-Archimedean Calabi–Yau theorem for Kähler manifolds.

2026 **Seminario di Geometria**, *Università degli studi di Roma Tor Vergata*, Rome, Italy

$\widehat{K}$-stability on Kähler manifolds

2026 **Workshop on K-stability**, East China Normal University, Shanghai, China

The non-Archimedean approach of the Yau–Tian–Donaldson conjecture

2026 **Geometry, HOlonomy and Structures with Torsion**, University of Münster, Münster, Germany
2026 **Geometric analysis seminar**, Westlake University, Hangzhou, China
2026 **Geometry and topology seminar series**, University of Science and Technology of China, Hefei, China
2025 **Séminaire de géométrie**, Centre de mathématiques Laurent-Schwartz, Palaiseau, France
2025 **Workshop on Singular canonical Kähler metrics on compact and non-compact manifolds**, Rényi Institute, Budapest, Hungary
2025 **Geometry seminar**, Max Plank Institute for Mathematics in the Sciences, Leipzig, Germany
2025 **Complex geometry seminar**, Mathematisches Institut, University of Münster, Munster, Germany
2025 **MARGE in Brest: Fibrations and Deformations**, Université de Bretagne Occidentale, Brest, France
2025 **Geometry Webinar AmSur /AmSul**, Webinar
2025 **Séminaire de Géométrie complexe**, Université Toulouse III - Paul Sabatier, Toulouse, France
2025 **Séminaire Géométries ICJ**, Université Claude Bernard Lyon 1, Lyon, France
2025 **Mathematics Seminar**, Aarhus University, Aarhus, Denmark
2024 **Séminaire de géométrie algébrique de Jussieu**, ENS, Paris, France
2024 **Séminaire systèmes dynamiques et géométrie**, Université d'Angers, Angers, France
2024 **Complex Analysis Seminar**, University of Bayreuth, Bayreuth, Germany
2024 **Geometric Analysis Seminar**, BICMR, Beijing, China
2024 **KASS Seminar**, Chalmers University, Gothenburg, Sweden
2024 **Algebraic Geometry Seminar**, University of Glasgow, Scotland
2024 **Any Complex Geometry Seminar**, University of California Berkeley, US

Other Info

Languages

Portuguese Native Speaker
Italian Native Speaker
English Fluent
Spanish Fluent
French Fluent

Memberships

2026 Member at the Société Mathématique de France